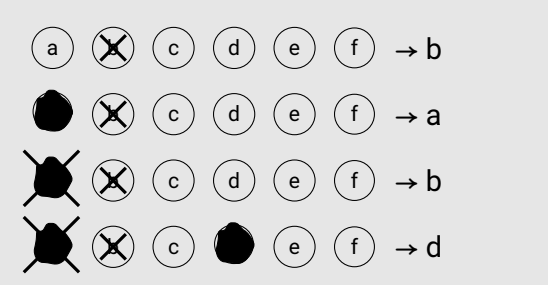
Exercises

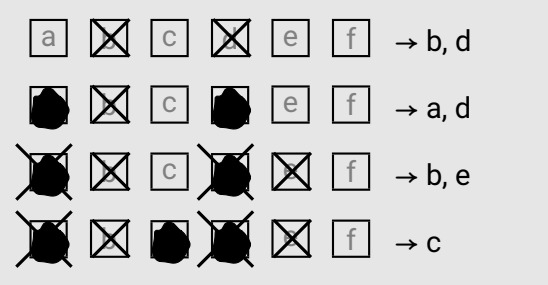
1	2	3	4

Surname, First name

1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9
0	0	0	0	0	0	0	0

Machine Learning 1 (52041MAL6Y)
Homework-based exam 2
8 Oct, 2026, 11:00-13:00





Answer multiple-choice questions as shown in the example. Circular checkboxes have only one correct answer. Square checkboxes may have multiple correct answers.

Pay attention to the following;

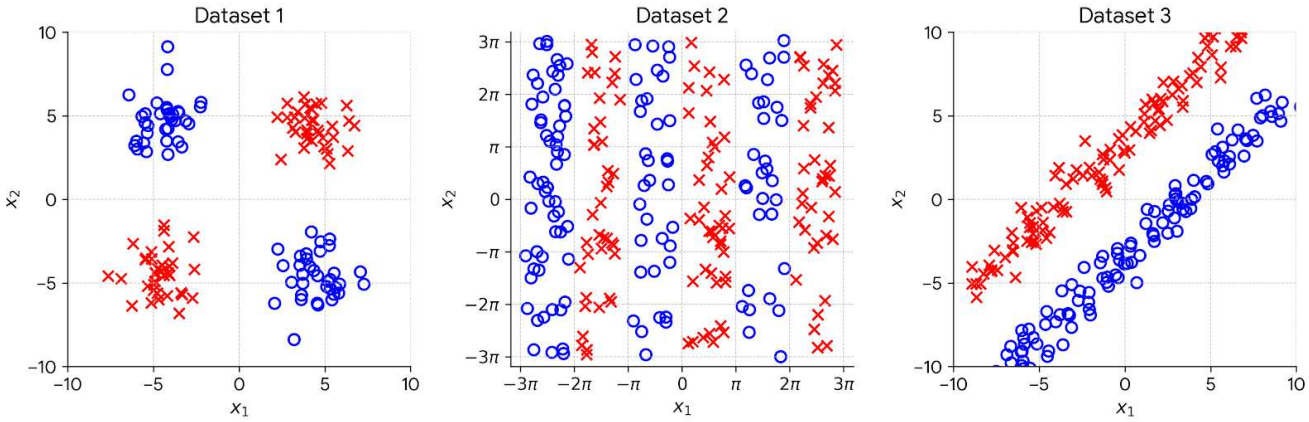
- **WRITE YOUR NAME AND STUDENT NUMBER ON THE FRONT PAGE** (don't forget to both write **and** mark the digits in the circles)
- Write all your answers in this exam booklet; it will be scanned and digitally graded.
  - Write your **answers inside the boxes** (it is ok to slightly go over the margins though).
  - The answer boxes should be large enough for your answer. However, **if you need to empty the answer box in order to start over, ask the invigilator for a blank sticker or use the *Emergency Box* at the end of the booklet.**
- You have **2 HOURS** to complete the exam (except for those with pre-approved extensions)
- In total **18 POINTS** can be earned divided over 3 exercises:
  - Exercise 1: Binary Classification [7p]
  - Exercise 2: Weighted Principal Component Analysis [3p]
  - Exercise 3: Mixture of Experts [8p]
- For open questions with derivations, you must include the procedure to get to your final solution. A correct solution with no explanation will not be awarded any points.
- We work with partial grading: you can get points even if you don't manage to solve the full question. Explaining your procedure clearly may earn you points even if your final calculation contains errors.

During the exam you are allowed to use:

- The provided cheat sheet.
- Basic calculator

[7p] Binary Classification

We consider the following datasets for binary classification:



Note: questions 1a and 1b, may have multiple correct answers, and you need to mark all of them for full credit.

0.5p **1a** Which dataset(s) can be separated using Logistic regression with linear features ($y = \sigma(\mathbf{w}^T\mathbf{x} + b)$)? *Note: with separated, we mean perfectly classified (100% training accuracy).*

- ☐ Dataset 1
- ☐ Dataset 2
- ☐ Dataset 3

0.5p **1b** And using logistic regression with nonlinear basis functions ($y = \sigma(\mathbf{w}^T\phi(\mathbf{x}))$)?

- ☐ Dataset 1
- ☐ Dataset 2
- ☐ Dataset 3

1p **1c** For each of the dataset(s) that you marked in **1b**, provide a suitable basis function $\phi(\mathbf{x})$ that makes the dataset separable via logistic regression.

1p **1d** In the context of classification, what does the Naive Bayes assumption consist of?

1p **1e** The Bayes-optimal classifier assigns $\mathbf{x}$ to the class with the highest posterior $p(C_k|\mathbf{x})$, $k \in \{1, 2\}$. Assuming equal class priors, $p(C_1) = p(C_2)$, show that this is the same as assigning $\mathbf{x}$ to the class with the highest class-conditional likelihood $p(\mathbf{x}|C_k)$.

3p **1f** Prove that no Naive Bayes classifier can perfectly classify **Dataset 1** (100% training accuracy), regardless of how the marginal densities $p(x_j|C_k)$ are chosen. You may assume equal class priors and use **1e**.
Hint: Consider the subset $\{(-5,5),(5,5),(5,-5),(-5,-5)\}$.

[3p] Weighted Principal Component Analysis

In the standard PCA setting, every data point is treated equally. However, in many real-world applications (e.g., sensor networks where some sensors are noisier than others), we might want to assign a confidence weight $w_i > 0$ to each data point $\boldsymbol{x}_i \in \mathbb{R}^d$.

Let $\bar{\boldsymbol{x}}_w = \frac{\sum_{i=1}^n w_i \boldsymbol{x}_i}{\sum_{i=1}^n w_i}$ be the weighted mean of the data. For convenience, we define the centered data points as $\boldsymbol{u}_i = \boldsymbol{x}_i - \bar{\boldsymbol{x}}_w$.

We wish to find an orthogonal projection matrix $P \in \mathbb{R}^{d \times d}$ of rank k that optimally represents this weighted dataset.
There are several ways to make "optimally represents" precise. In this exercise we consider two formulations, **[A]** and **[B]**, and show that they lead to the same projection. You do not need to solve these optimization problems: the questions only ask you to rewrite and compare them, and none of them require derivatives.

[A] Maximize the Weighted Scatter:

$$\max_P \sum_{i=1}^n w_i \|P\boldsymbol{u}_i\|^2$$

[B] Minimize the Weighted Reconstruction Error:

$$\min_P \sum_{i=1}^n w_i \|\boldsymbol{u}_i - P\boldsymbol{u}_i\|^2$$

2p **2a** Prove that formulation **[A]** is equivalent to $\max_P \mathrm{Tr}(S_w P)$. Explicitly define the matrix S_w in your proof.
Hint: Consider the cyclic property of the trace: $tr(ABC) = tr(BCA) = tr(CAB)$

↳

1p **2b** Formulation **[B]** is equivalent to $\min_P \mathrm{Tr}(S_w(I-P))$. Given this result (which you don't need to prove), explain why formulations **[A]** and **[B]** share the same optimal projection matrix P.

[8p] Mixture of Experts

Consider K expert models for a dataset of N data points $\{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)\}$, where $\mathbf{x}_n \in \mathbb{R}^D$ and y_n is the ground-truth label for $\mathbf{x}_n$. Let z_n denote a categorical random variable indicating which expert model is active. Let $\boldsymbol{\Theta} \in \mathbb{R}^{D \times K}$ contain the expert parameter vectors as columns.
Assume each $y_n \in \{0, 1\}$ and follows a Bernoulli distribution. Expert k has the predictive model

$$p(y_n \mid \mathbf{x}_n, z_n = k; \boldsymbol{\theta}_k) = \text{Bernoulli}\left(y_n \mid \rho_{nk}\right),$$

where

$$\rho_{nk} = \sigma(\boldsymbol{\theta}_k^T \mathbf{x}_n), \qquad \sigma(a) = \frac{1}{1 + \exp(-a)},$$

and

$$\text{Bernoulli}(y \mid \rho) = \rho^y (1 - \rho)^{1-y}, \qquad y \in \{0, 1\}.$$

The routing mechanism determines how relevant each expert is for a specific datapoint:

$$p(z_n = k \mid \mathbf{x}_n; \boldsymbol{\Phi}) = \pi_{nk} = \frac{\exp(\boldsymbol{\phi}_k^T \mathbf{x}_n)}{\sum_{j=1}^{K} \exp(\boldsymbol{\phi}_j^T \mathbf{x}_n)},$$

where $\boldsymbol{\Phi} = [\boldsymbol{\phi}_1, \ldots, \boldsymbol{\phi}_K] \in \mathbb{R}^{D \times K}$.

1.5p **3a** Since we have no prior information about which expert is responsible for a particular prediction, marginalize over all possible experts. Write $p(y_n \mid \mathbf{x}_n; \boldsymbol{\Theta}, \boldsymbol{\Phi})$ as a marginalization over z_n. Then, assuming i.i.d data, write down the likelihood $p(\mathbf{y} \mid \mathbf{X}; \boldsymbol{\Theta}, \boldsymbol{\Phi})$ and the log-likelihood $\mathcal{L}$ of the entire dataset.

1p **3b** Write down the expression for the responsibility $r_{ni} \coloneqq p(z_n = i \mid \mathbf{x}_n, y_n; \boldsymbol{\Theta}, \boldsymbol{\Phi})$ in terms of the routing probabilities $p(z_n = k \mid \mathbf{x}_n; \boldsymbol{\Phi})$ and the expert-conditional probabilities $p(y_n \mid \mathbf{x}_n, z_n = k; \boldsymbol{\theta}_k)$.

2p **3c** Show that the derivatives of the log likelihood $\mathcal{L}$ with respect to the parameters are

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}_i} = \sum_{n=1}^{N} r_{ni} \frac{\partial}{\partial \boldsymbol{\theta}_i} \log p(y_n \mid \mathbf{x}_n, z_n = i; \boldsymbol{\theta}_i) \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial \boldsymbol{\phi}_i} = \sum_{n=1}^{N} \sum_{k=1}^{K} r_{nk} \frac{\partial}{\partial \boldsymbol{\phi}_i} \log p(z_n = k \mid \mathbf{x}_n; \boldsymbol{\Phi}).$$

2p **3d** Insert the explicit expressions for the probability distributions into the derivatives given in 3c to get the explicit form of the derivatives of the log likelihood $\mathcal{L}$ with respect to θ_i and ϕ_i.

1.5p **3e** Write down (pseudo-code for) an iterative algorithm that maximizes the log-likelihood by jointly optimizing $\boldsymbol{\Theta}$ and $\boldsymbol{\Phi}$. Include: (i) initialization, (ii) what is recomputed each iteration and the update rules and (iii) a convergence criterion. You may refer to the gradients symbolically, e.g. $\frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}_i}$.

Emergency Box

4 *Use this space only as a last resort* to provide answer to a previous question that didn't fit in the corresponding box. Indicate which question it is about (and also indicate it in the answer box of the corresponding question).

↳

